Better to Ask in English: Evaluation of Large Language Models on English, Low-resource and Cross-Lingual Settings

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Abstract

Large Language Models (LLMs) are trained on massive amounts of data, enabling their application across diverse domains and tasks. Despite their remarkable performance, most LLMs are developed and evaluated primarily in English. Recently, a few multi-lingual LLMs have emerged, but their performance in low-resource languages, especially the most spoken languages in South Asia, is less explored. To address this gap, in this study, we evaluate LLMs such as GPT-4, Llama 2, and Gemini to analyze their effectiveness in English compared to other low-resource languages from South Asia (e.g., Bangla, Hindi, and Urdu). Specifically, we utilized zero-shot prompting and five different prompt settings to extensively investigate the effectiveness of the LLMs in cross-lingual translated prompts. The findings of the study suggest that GPT-4 outperformed Llama 2 and Gemini in all five prompt settings and across all languages. Moreover, all three of the LLMs performed better for English language prompts than other low-resource language prompts. This study extensively investigates LLMs in low-resource language contexts to highlight the improvements required in LLMs and language-specific resources to develop more generally purposed NLP applications.

1 Introduction

Large Language Models (LLMs) have recently undergone significant advancements and have transformed the landscape of Natural Language Processing (NLP). LLMs are trained using large datasets, which enable them to recognize, translate, predict, or generate text and other content (Yang et al., 2023). Recent advances in LLMs enable the development of more powerful, efficient, and versatile NLP systems with broad applicability across different domains and tasks (Zhao et al., 2023).

Researchers have been evaluating LLMs for several NLP tasks, including machine translation (Xu et al., 2023; Lyu et al., 2023), text summarization (Pu et al., 2023; Zhang et al., 2023a), reasoning (Suzgun et al., 2022; Miao et al., 2023), and mathematics (Lu et al., 2023; Rane, 2023). However, most of these works, which utilized LLMs in these domains, are limited to high-resource languages such as English (Xing, 2024). Nonetheless, these works laid the foundation for exploring LLMs for downstream tasks in low-resource languages.

On the other hand, the prevailing literature predominantly employs traditional machine learning or transformer-based models for analyzing low-resource languages (Hedderich et al., 2020). Recently, a few multi-lingual LLMs have emerged (Asai et al., 2023), but their performance on low-resource languages, especially the most spoken languages in South Asia, is less explored. Three of the most widely spoken languages in South Asia are Hindi, with 610 million speakers; Bangla, with 273 million speakers; and Urdu, with 232 million speakers. Moreover, Hindi is the 3rd, Bangla is the 7th, and Urdu is the 10th most spoken language in the world. Despite collectively representing over 1 billion speakers, computational resources for these three languages remain limited. In this study, we explore the effectiveness of LLMs in Bangla, Hindi, and Urdu languages vs English[1].

Very few works in low-resource languages utilize LLMs to solve Natural Language Processing (NLP) tasks, and most of the work is limited to some extent. A very few noteworthy works employ LLMs in Bangla (Liu et al., 2023; Hasan et al., 2023b), Urdu (Koto et al.,

[1]We chose these languages because the authors are native speakers of Bangla, Hindi, and Urdu.

2024), and Hindi (Kumar and Albuquerque, 2021; Koto et al., 2024). These studies show that LLMs can achieve identical and better (in some cases) results compared to transformer-based models and traditional machine-learning techniques. However, more intensive studies are required to evaluate LLMs in these low-resource languages to determine their effectiveness and suggest potential improvement required in LLMs.

To address this gap, in this study, we analyze various LLMs (e.g., GPT-4, Llama 2, and Gemini) with zero-shot learning across both English and low-resource languages (e.g., Bangla, Hindi, and Urdu) under different prompt settings [2]. Specifically, we evaluate their effectiveness in English compared to other low-resource languages from South Asia. We perform cross-lingual prompt translation from originally designed prompts to develop various prompt settings and investigate the performance of LLMs under those prompt settings. Our results show that GPT-4 outperformed Llama 2 and Gemini in all languages. Furthermore, Llama 2 cannot recognize prompts in low-resource languages, resulting in the lowest performance. Additionally, LLMs perform better in English than in other languages such as Bangla, Hindi, and Urdu. The key contributions of this work include:

- We investigate the effectiveness of zero-shot prompting with LLMs (GPT-4, Llama 2, Gemini) for cross-lingual tasks involving low-resource languages from South Asia (Bangla, Hindi, Urdu). The findings demonstrate that the translation quality of prompts has minimal impact on LLM performance, suggesting potential for language-agnostic prompting techniques in low-resource NLP applications.

- Our work explores five unique prompt settings to analyze the influence of prompt design on LLM performance in high-resource (English) and low-resource languages. The results show that English prompts consistently outperform prompts in other languages, even with translated prompts.

- We present a novel approach incorporating Natural Language Instructions (NLI) and zero-shot prompting. Providing task descriptions and expected outputs through NLI creates a richer context for LLMs, improving performance in English and low-resource language tasks.

- We conduct a comparative analysis of 3 prominent LLMs (GPT-4, Llama 2, Gemini) to evaluate their effectiveness in low-resource languages. The findings reveal that GPT-4 outperforms the other 2 models across all languages and prompt settings.

[2]Generic LLM names are used throughout the paper, exact versions are GPT-4, Llama-2–70b-chat, and Gemini Pro

2 Related Works

The emergence of LLMs marks a major milestone in NLP, proficient in various tasks such as language understanding (Dou et al., 2019), text generation (Huang et al., 2020), question answering (Giampiccolo et al., 2007), and sentiment analysis (Yu et al., 2018). However, their performance in low-resource languages needs improvement (Robinson et al., 2023).

2.1 LLM for English

In recent years, extensive resources and benchmarks for English (Wang et al., 2018; Williams et al., 2017) have fueled the development of LLMs. LLMs significantly impact tasks such as question answering (Akter et al., 2023; Tan et al., 2023; Zhuang et al., 2023), reasoning (Suzgun et al., 2022; Miao et al., 2023), and machine translation (Xu et al., 2023; Lyu et al., 2023). They are also applied in NLI (Gubelmann et al., 2023), Sentiment Analysis (Sun et al., 2023; Zhang et al., 2023b), and Hate Speech Detection (Zhang et al., 2024). Multilingual LLMs show strong performance across languages (Sitaram et al., 2023), yet struggle with low-resource languages (Ahuja et al., 2023). However, LLMs are known for their capabilities to understand the relationships among text sequences and produce results similar to state-of-the-art techniques (Pahwa and Pahwa, 2023; Gubelmann et al., 2023). The study of Brown et al. (2020) shows that LLMs can perform various tasks by following instructions and learning from just a few examples provided in context. Moreover, English-centric LLMs such as Llama 2 can almost perfectly

match the input and output of the language when adjusted with few multilingual conversational instructions, despite their limited exposure to other languages (Kew et al., 2023). The study of Asai et al. (2023) focuses on few-shot learning and instruction fine-tuning of smaller LLMs (such as mT5, mT0) and ChatGPT to improve the tasks (such as NLI, question answering, sentiment, commonsense reasoning, etc.) performance.

2.2 LLM for Low-resource Languages

Research and practical applications in the field of NLP are centered around high-resource languages that typically have large annotated corpora, well-established tools and libraries, and robust language models trained on large data. Despite being resourceful, the researchers prioritize analyzing high-resource languages and overlook other languages spoken by billions of people (Bender, 2019).

Most of the work across several downstream tasks in low-resource languages employs traditional machine learning or transformer-based language models (Jahan and Oussalah, 2023; Chhabra and Vishwakarma, 2023). To improve the Natural Language Interface (NLI) task across low-resource languages, several researchers have attempted to develop benchmark datasets and frameworks (Bhattacharjee et al., 2021; Rahman et al., 2017; Chakravarthy et al., 2020). Similar attempts were also made in tasks such as Sentiment Analysis (Islam et al., 2021; Hasan et al., 2020a, 2023a; Patra et al., 2018; Muhammad and Burney, 2023), Hate Speech Detection(Badjatiya et al., 2017; Zimmerman et al., 2018), etc. These works do not explore the use of LLMs for those tasks above.

Researchers have recently been shifting their attention towards developing applications for low-resource languages. As a result, several authors are evaluating LLMs in low-resource languages across several tasks (Liu et al., 2023; Hasan et al., 2023b; Kabir et al., 2023; Koto et al., 2024; Kumar and Albuquerque, 2021; Hee et al., 2024; García-Díaz et al., 2023). However, there are several challenges in evaluating LLMs in low-resource languages for different tasks (Ahuja et al., 2023; Chung et al., 2023). Researchers have developed low-resource language benchmark (Asai et al., 2023) and datasets (Aggarwal et al., 2022; Khan et al., 2024) to address such challenges. INDICXNLI (Aggarwal et al., 2022), INDICLLMSUITE (Khan et al., 2024) dataset consists of several Indian regional languages but do not have any resources in Urdu. The benchmark BUFFET (Asai et al., 2023) incorporates most spoken South Asian languages in NLI tasks; however, they do not incorporate Bangla in their experiments. Along with the several low-resource benchmarks and datasets, many prompting techniques were also explored to evaluate LLMs on low-resource languages (Qin et al., 2023; Huang et al., 2023).

Recently, a few multi-lingual LLMs have emerged, but their performance in low-resource languages, especially the most spoken languages in South Asia, is less explored. To address this gap, in this study, we evaluate LLMs such as GPT-4, Llama 2, and Gemini to analyze their effectiveness in English compared to other low-resource languages from South Asia (e.g., Bangla, Hindi, and Urdu).

3 Methodology

3.1 Prompt Approach

The quality of the prompt impacts the performance of LLMs (White et al., 2023). A well-designed prompt helps users and developers leverage the capabilities of LLMs to retrieve information that aligns with their specific interests. LLMs are usually trained on diverse datasets. Thus, providing clear instructions on interacting with LLMs and producing the desired information is essential. Designing a good prompt is an iterative process that requires instructions refining through successive interactions, enabling the LLMs to learn and produce desired results. In this study, we have employed zero-shot prompting and provided instruction in natural language. To help LLMs generate more relevant output, we provided detailed instructions containing the task descriptions and prompt template in Table 1. We used the same instruction format for all languages and LLMs.

Furthermore, to leverage the ability of GPT-4 to receive role information and perform according to that specific role, we provided role information along with the prompt. During the experiment, Gemini Pro blocked most prompts

for containing harmful and inappropriate texts and made no predictions. To obtain the predictions for those harmful contents, we changed the safety settings of the Gemini Pro models. Despite changing the safety setting, models still did not make any predictions. See Appendix A for safety settings.

3.2 Experimental Details

3.2.1 Data

In this study, we employed LLMs to evaluate the Natural Language Interface (NLI) and Classification task due to the absence of cross-lingual datasets for other downstream NLP tasks.

We used the most widely known and publicly available cross-lingual natural language inference (XNLI) dataset developed by (Conneau et al., 2018). The dataset contains 15 languages, including low-resource languages such as Urdu and Hindi. English validation and test sets of MultiNLI were manually translated, while the train set was machine-translated for all languages. The dataset contains 392,702 train, 2,490 validation, and 5,010 test samples. Each data sample contains the premise, hypothesis, and corresponding labels (e.g., entailment, neutral, and contradiction). In our study, we selected test sets of English, Hindi, and Urdu languages from the XNLI dataset. We also selected a Bangla XNLI dataset available for public use, and was generated by translating the XNLI dataset using English to Bangla translator model (Hasan et al., 2020b), which is widely used and the only large-scale study on English-to-Bangla machine translation. The Bangla XNLI dataset contains 381,449 train, 2,419 validation, and 4,895 test samples.

The most spoken languages in South Asia—Hindi, Bangla, and Urdu—are used daily by approximately 1.1 billion people [3]. Despite their widespread use, these languages are considered low-resource in NLP. We included all three in the XNLI dataset to assess LLMs' performance compared to English. Having at least one native speaker of these languages among the authors aids in organizing and understanding LLMs' responses during experiments.

[3] https://en.wikipedia.org/wiki/List_of_languages_by_total_number_of_speakers

3.2.2 Prompt Settings

Table 2 shows the prompt settings and prompts of all languages used in our study. Setting P1 consists of four language-specific prompts, and each of the four prompts was designed by 4 native speakers of Bangla, English, Hindi, and Urdu languages. Setting P2, P3, P4, and P5 was created by translating the prompt from English, Bangla, Urdu, and Hindi respectively. Classes names such as 'Entailment', 'Contradiction', and 'Neutral' were used consistently for all language prompts [4]. We used Google Translatorfor translating prompts across four languages.

3.2.3 Post-Processing

Our designed prompt instructed the LLMs to output the class labels (e.g., entailment, neutral, and contradiction) as the response. However, LLMs often returned extra characters and words along with the label. We processed the outputs by removing unknown characters and filtering the class labels using regular expressions to handle such responses. When the LLM response contained no class label, we classified them as "None" to get the final results.

3.2.4 Evaluation Metrics

To evaluate the performances, we computed accuracy, precision, recall, and F1 scores for all the experimental settings to evaluate the effectiveness of LLMs on cross-lingual translated prompts. We have computed weighted precision, recall, and macro F1 scores to deal with class imbalance.

4 Results and Discussion

In this section, we present and discuss the experimental results of our study. Firstly, we discuss the performance of LLMs in English compared to low-resource languages (i.e., Bangla, Hindi, and Urdu). Then, we provide a comparison of LLMs across different settings. Finally, we delve into the results of each setting (e.g., P1, P2, P3, P4, and P5). Additionally, we provide class-wise precision and recall scores in Table 5.

4.1 English vs Low-resource Languages

The experimental results presented in Table 3 indicate that all LLMs exhibit supe-

[4] Class labels are in English for all languages

Model	Prompt Template

GPT-4	[{
	'role': 'user',
	'content': "Classify the following 'premise' and 'hypothesis' into one of the following classes: 'Entailment', 'Contradiction', or 'Neutral'. Provide only label as your response."
	premise: [PREMISE_TEXT]
	hypothesis: [HYPOTHESIS_TEXT]
	label:
	},
	{
	role: 'system',
	content: "You are an expert data annotator and your task is to analyze the text and find the appropriate output that is defined in the user content."
	}]

Llama 2 and Gemini	Classify the following 'premise' and 'hypothesis' into one of the following classes: 'Entailment', 'Contradiction', or 'Neutral'. Provide only label as your response. premise: [PREMISE_TEXT] hypothesis: [HYPOTHESIS_TEXT] label:

Table 1: Prompts used for zero-shot learning

rior performance for English prompts (indicated by Underline) across the prompt settings. Note that P2 setting does not contain English prompts. In the P1 setting (original prompts), GPT-4 outperforms other languages in English by 17.75%, 14.82%, and 22.02% for Bangla, Hindi, and Urdu respectively. This trend persists across all settings, with English consistently demonstrating superior performance compared to other languages, even with translated prompts. In settings P3 and P4, GPT-4 exhibits enhanced performance in English compared to Hindi and Urdu by 18.28% and 21.60%, respectively. Similarly, in setting P5, English outperforms Bangla and Urdu by 10.81% and 10.33%, respectively. Although Bangla and Hindi perform better than Urdu, there remains a significant performance gap compared to English. However, the performance differences among low-resource languages across all settings are negligible.

Similarly, Llama 2 performs better in English but poorly in other languages. The performance gap between English and other languages is substantial, with English outperforming Bangla, Hindi, and Urdu by 22.03%, 20.19%, and 23.20%, respectively. This trend persists across all settings, with English consistently performing better than other languages.

In contrast, the performance disparity between English and other languages is slightly lower in Gemini Pro compared to GPT-4 and Llama 2. Gemini Pro performs better in English than Bangla, Hindi, and Urdu by 11.62%, 11.64%, and 27.25%, respectively, across all settings. This trend continues in translated prompt settings (P3, P4, and P5), with English consistently outperforming other languages.

Such bias towards English in LLMs can be attributed to several factors. One significant reason is the disproportionate training distribution, with most LLMs predominantly trained on extensive English corpora. For instance, in Llama 2, approximately 90% of the training data is sourced from English, leaving only 10% representing other languages worldwide. This skewed training distribution significantly contributes to the observed biases towards low-resource languages. The findings underscore the need for substantial improvements in LLMs to ensure better generalization across low-resource languages.

4.2 Comparison among LLMs

In this section, we compare the performance of different LLMs. Results showed that GPT4 and Llama 2 could make predictions for all data samples, while Gemini Pro did not make any predictions for prompts containing harmful content despite adjusting the safety setting. The number of unpredicted samples was very low (ranging from 1-4) and negligible. Additionally, LLMs returned unknown characters and words alongside class labels, which we addressed during post-processing (as described in 3.2.3). We assigned an inverse class to sam-

	P1 Settings
BN	নিম্নলিখিত 'premise' and "hypothesis" কে 'entailment', or 'contradiction', or 'neutral' এর যে কোনো একটি শ্রেণীতে বিভক্ত করো । উত্তর হিসাবে শুধু শ্রেণী দিবা।
EN	Classify the following ‘premise’ and “hypothesis” into one of the following classes: ‘entailment’, or ‘contradiction’, or ‘neutral’. Provide only class as your response.
HI	निम्नलिखित 'premise' और "hypothesis" को निम्नलिखित श्रेणियों में से एक में वर्गीकृत करें: 'entailment', या 'contradiction', या 'neutral' आपकी प्रतिक्रिया केवल एक श्रेणी के रूप में होनी चाहिए।
UR	'entailment', or 'contradiction' کو درج ذیل:'Premise' اور 'hypothesis' or 'neutral' میں سے کسی ایک کلاس میں درجہ بندی کریں اپنے جواب میں صرف کلاس فراهم کریں۔
	P2 Settings
BN	নিম্নলিখিত 'premise' এবং 'hypothesis' কে নিম্নলিখিত শ্রেণীগুলির মধ্যে একটিতে শ্রেণীবদ্ধ করুন: 'entailment', বা 'contradiction', বা 'neutral'। আপনার প্রতিক্রিয়া হিসাবে শুধুমাত্র শ্রেণী প্রদান করুন ।
HI	निम्नलिखित 'premise' और 'hypothesis' को निम्नलिखित वर्गों में से एक में वर्गीकृत करें: 'entailment', या 'contradiction', या 'neutral'। अपनी प्रतिक्रिया के रूप में केवल कक्षा प्रदान करें।
UR	درج ذیل 'premise' اور 'hypothesis' کو درج ذیل میں سے کسی ایک طبقے میں درجہ بندی کریں: 'entailment' ، یا "contradiction"، یا 'neutral'۔ اپنے جواب کے طور پر صرف کلاس فراهم کریں۔
	P3 Settings
EN	Classify the following ‘premise’ and ‘hypothesis’ into any one of the categories ‘entailment’, or ‘contradiction’, or ‘neutral’. Just give category as answer.
HI	निम्नलिखित 'premise' और hypothesis' को 'entailment', या 'Contradiction', या 'neutral' में से किसी एक श्रेणी में वर्गीकृत करें। उत्तर के रूप में केवल श्रेणी दें।
UR	مندرجہ ذیل 'premise' اور 'hypothesis' کو زمرہ جات میں سے کسی ایک میں درجہ بندی کریں 'entailment'، 'neutral'۔ صرف جواب کے طور پر زمرہ دیں ، یا 'contradiction' یا
	P4 Settings
BN	`premise' বা 'hypothesis'কে নিম্নলিখিত শ্রেণীগুলির মধ্যে একটিতে শ্রেণীবদ্ধ করুন: 'entailment' এবং 'contradiction' বা 'neutral'। আপনার উত্তরে শুধুমাত্র ক্লাস প্রদান করুন।
EN	Classify ‘premise’, or ‘hypothesis’ into one of the following classes: ‘entailment’ and ‘contradiction’ or ‘neutral’. Provide only the class in your answer.
HI	'premise', या 'hypothesis' को निम्नलिखित वर्गों में से एक में वर्गीकृत करें: 'entailment' और 'contradiction' या 'neutral'। अपने उत्तर में केवल वर्ग प्रदान करें।
	P5 Settings
BN	নিম্নলিখিত 'premise' এবং 'hypothesis'কে নিম্নলিখিত বিভাগগুলির মধ্যে একটিতে শ্রেণীবদ্ধ করুন: 'entailment', বা 'contradiction', বা 'neutral' আপনার প্রতিক্রিয়া শুধুমাত্র একটি বিভাগে হওয়া উচিত।
EN	Classify the following ‘premise’ and ‘hypothesis’ into one of the following categories: ‘entailment’, ‘contradiction’, or ‘neutral’ Your response should be in one category only.
UR	مندرجہ ذیل 'premise' اور "hypothesis" کو مندرجہ ذیل مضامین میں سے ایک میں زمرہ بندی کریں: 'entailment' ، آپ کا ردعمل صرف ایک زمرے کے طور پر ہونا چاهیے۔ 'neutral' ، یا 'contradiction' یا

Table 2: Settings **(P1)** represents prompts provided by native speakers. **P2 Settings, P3 Settings, P4 Settings, and P5 Settings** represents the translated prompts from English, Bangla, Urdu, and Hindi respectively. BN: Bangla, EN: English, HI: Hindi, and UR: Urdu.

ples with invalid labels, reducing the overall performance of LLMs during evaluation metric calculations. The total number of invalid labels returned by LLMs is shown in Table 4.

Model	Lang.	P1		P2		P3		P4		P5	
		Acc.	$F1_{macro}$	Acc.	$F1_{macro}$	Acc.	$F1_{macro}$	Acc.	$F1_{macro}$	Acc.	$F1_{macro}$
GPT-4	**BN**	68.72	**69.05**	68.72	**69.05**	-	-	70.73	**71.18**	70.26	**70.54**
	EN	86.73	**86.79**	-	-	87.03	**87.08**	82.42	**81.99**	86.73	**86.81**
	HI	71.52	**71.97**	70.26	**70.73**	68.52	**68.80**	71.20	**71.67**	-	-
	UR	65.07	**64.77**	66.45	**66.73**	65.31	**65.48**	-	-	66.77	**66.95**
Llama 2	**BN**	35.49	36.73	33.24	31.27	-	-	35.42	35.91	32.05	33.59
	EN	65.73	58.76	-	-	60.80	56.78	62.59	59.17	63.59	60.11
	HI	36.27	38.57	33.53	33.66	38.02	40.89	33.19	33.43	-	-
	UR	36.39	35.56	39.48	35.83	36.77	37.65	-	-	33.31	36.35
Gemini	**BN**	62.19	61.74	60.56	60.16	-	-	56.56	52.41	60.67	60.30
	EN	73.71	73.36	-	-	73.65	73.09	62.94	60.73	74.71	74.72
	HI	61.90	61.72	61.88	61.70	62.10	61.47	52.10	48.19	-	-
	UR	49.75	46.11	57.30	56.47	59.04	58.40	-	-	57.25	56.34

Table 3: Performances of the LLMs across the settings, and languages. Underline shows the best F1 score among four languages across all settings and LLMs. Blue indicates the best F1 score in all settings among three LLMs. **Bold** represents the highest F1 score among LLMs in all five prompt settings for each of the four languages. Lang.: Language, Acc.: Accuracy, BN: Bangla, EN: English, HI: Hindi, and UR: Urdu

Model	Lang.	P1	P2	P3	P4	P5
GPT-4	**BN**	1	13	-	2	48
	EN	0	-	0	0	0
	HI	9	13	17	7	-
	UR	43	53	18	-	130
Llama 2	**BN**	4147	1678	-	2195	2827
	EN	19	-	619	322	588
	HI	4010	4966	3306	4923	-
	UR	1308	790	1917	-	3872
Gemini	**BN**	142	149	-	68	127
	EN	138	-	143	80	129
	HI	111	131	138	133	-
	UR	1330	105	85	-	77

Table 4: Number of invalid labels returned by LLMs across the settings, models, and languages. Lang.: language.

Table 3 demonstrates that GPT-4 consistently outperforms Llama 2 and Gemini Pro across all five settings (indicated by Blue). Further investigation revealed that GPT -4 accurately understood prompts and provided appropriate responses, with minimal unwanted characters and words accompanying the actual labels. Interestingly, GPT-4 showed superior performance in English prompts across all settings, suggesting its robustness in understanding English language prompts compared to others. Moreover, GPT-4's multi-lingual capabilities improved performance in languages such as Bangla, Hindi, and Urdu.

Conversely, Gemini Pro performed better than Llama 2 but not as well as GPT-4. Despite its performance, Gemini Pro returned many unwanted characters, words, or invalid labels, which affected its overall performance. Additionally, Gemini Pro struggled with predicting samples containing harmful content and lacked support for the Urdu language, further limiting its performance.

In contrast, Llama 2 exhibited poor performance compared to GPT-4 and Gemini Pro. Trained on 90% English data, Llama 2 faced challenges in understanding prompts in low-resource languages such as Bangla, Hindi, and Urdu. Additionally, Llama 2 returned more unwanted characters, words, or invalid labels, significantly impacting its performance during evaluation metric calculations. Moreover, Llama 2 showed dominance in predicting the *neutral* class over *entailment* and *contradiction* classes, indicating its difficulty in accurately predicting sentences belonging to these classes.

In summary, GPT-4 demonstrated superior performance to Llama 2 and Gemini Pro across all settings and languages, showcasing its accuracy in understanding prompts and providing appropriate responses. Gemini Pro performed better than Llama 2 but lacked support for Urdu and struggled with predicting samples containing harmful content. Conversely, Llama 2 faced challenges in understanding prompts in low-resource languages and exhibited dominance in predicting the *neutral* class. These findings suggest that GPT-4's multi-lingual ca-

Model	Lang	P1			P2			P3			P4			P5		
		Cont.	Ent.	Neut.	Cont.	Ent.	Neut.	Cont.	Ent.	Neut.	Cont.	Ent.	Neut.	Cont.	Ent.	Neut.
GPT-4	**BN**	73.51	66.72	66.91	73.51	66.72	66.91	-	-	-	76.81	69.93	66.80	74.92	68.59	68.11
	EN	90.90	87.56	81.92	-	-	-	91.22	87.84	82.17	89.83	83.44	72.71	90.91	87.72	81.80
	HI	77.85	69.81	68.24	76.97	67.96	67.26	76.06	63.88	66.46	78.30	69.62	67.08	-	-	-
	UR	73.20	61.74	59.37	72.68	63.47	64.03	71.57	61.04	63.82	-	-	-	73.47	63.32	64.06
Llama 2	**BN**	06.39	11.91	91.88	09.30	39.02	45.48	-	-	-	10.90	35.74	61.08	03.70	28.87	68.19
	EN	88.36	68.11	19.79	-	-	-	76.77	61.38	32.20	82.97	62.59	31.95	80.62	63.16	36.56
	HI	11.84	14.38	89.48	0.60	0.77	99.61	18.29	24.40	79.99	00.90	00.66	98.02	-	-	-
	UR	29.61	40.41	36.65	39.02	45.60	22.86	24.32	37.55	51.09	-	-	-	11.00	12.64	85.42
Gemini	**BN**	70.08	59.88	55.25	69.03	54.74	56.72	-	-	-	68.98	30.09	58.17	68.63	53.30	58.98
	EN	82.05	72.53	65.48	-	-	-	80.61	74.51	64.14	79.23	42.43	60.53	82.53	73.92	67.72
	HI	70.37	57.10	57.68	69.67	56.08	59.34	70.57	61.40	52.43	63.41	25.01	56.15	-	-	-
	UR	53.75	20.95	63.62	67.05	50.56	51.79	68.84	50.56	55.80	-	-	-	66.62	51.16	51.24

Table 5: Class-wise $F1_{macro}$ score for GPT-4, Llama 2, and Gemini across five prompt settings. Lang.: Language, BN: Bangla, EN: English, HI: Hindi, and UR: Urdu, Cont: contradiction, Ent: Entailment, Neut: Neutral.

pabilities and precise, prompt understanding contribute to its effectiveness in various language settings.

4.3 Cross-Lingual Translated Prompts

This section discusses the results of LLMs across languages in each setting. To make valid comparisons among different settings (i.e., P1, P2, P3, P4, and P5), we only consider the best F1 score achieved among the three LLMs (i.e., GPT-4, Llama 2, and Gemini Pro) for each language. Table 3 shows the best scores for each language for individual prompt settings (indicated by Blue).

For the original prompts in the P1 setting, the highest F1 scores for each language are Bangla (69.05%), English (86.79%), Hindi (71.97%), and Urdu (64.77%). In P2, the highest F1 scores for each language are Bangla (69.05%), Hindi (70.73%), and Urdu (66.73%). For setting P3, the highest F1 scores for each language are English (87.08%), Hindi (68.80%), and Urdu (65.48%). In setting P4, the highest F1 scores for each language are Bangla (71.18%), English (81.99%), and Hindi (71.67%). Finally, for setting P5, the highest F1 scores for each language are Bangla (86.81%), English (70.54%), and Urdu (66.95%).

We also observe that the Hindi-to-Bangla translated prompt in setting P5 achieved a higher F1 score (86.81%) than the original Bangla prompt in setting P1 (69.05%). Similarly, for the Urdu language, the translated prompt in setting P5 achieved a higher F1 score (66.95%) than the original Urdu prompt. For the English language, the Bangla-to-English translated prompt in setting P3 achieved a higher F1 score (87.08%) than the original English prompt. However, the original prompt achieved the highest F1 score for the Hindi language compared to other translated prompts. Our results suggest that the translation of a specific language does not significantly affect the prediction capabilities of LLMs.

5 Conclusion and Future Work

In this study, we evaluated LLMs including GPT-4, Llama 2, and Gemini Pro across English and low-resource languages such as Bangla, Hindi, and Urdu, focusing on the NLI task due to dataset limitations. Despite recent advancements in LLMs, their performance on low-resource languages, particularly those spoken in South Asian countries, remains underexplored. Utilizing zero-shot prompting techniques in five different settings, our findings reveal GPT-4 consistently outperforms Llama 2 and Gemini Pro, with Gemini Pro showing better performance than Llama 2, particularly struggling with low-resource language prompts. Interestingly, English prompts yield superior responses compared to low-resource language prompts, and translating prompts from one language to another occasionally enhances performance. Future research should prioritize collecting low-resource datasets and developing resources for South Asian languages to improve LLM generalization, while incorporating techniques like few-shot prompting and developing cross-lingual datasets for various tasks would facilitate comprehensive evaluations on low-resource languages.

Limitations

We evaluated LLMs on a single dataset (i.e., XNLI) and a single task (i.e., NLI), which may not accurately reflect the capabilities of LLMs across a broader range of tasks and datasets. The unavailability of cross-lingual datasets such as XNLI restricted us from conducting our study on a broader scale, incorporating more tasks and datasets. Additionally, we only utilized zero-shot prompting techniques and did not explore explicit prompting techniques (e.g., few-shot prompting) to enhance the performance. The utilization of only prompting techniques may limit the generalization of our findings. However, we could not explore other prompting techniques due to resource limitations; conducting experiments with LLMs requires premium access, which comes at a hefty cost.

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A Prompt and Safety Settings

In this section, we present the safety settings in Table 6 that are used for Gemini Pro across all prompt settings to prevent it from blocking predictions for harmful content.

Category	Threshold
HARM_CATEGORY_HARASSMENT	BLOCK_NONE
HARM_CATEGORY_HATE_SPEECH	BLOCK_NONE
HARM_CATEGORY_SEXUALLY_EXPLICIT	BLOCK_NONE
HARM_CATEGORY_DANGEROUS_CONTENT	BLOCK_NONE
HARM_CATEGORY_SEXUAL	BLOCK_NONE
HARM_CATEGORY_DANGEROUS	BLOCK_NONE

Table 6: Safety setting used for Gemini Pro model to prevent blocking the predictions for harmful content.

B Detailed Experimental Results

Table 7 shows class-wise experimental results (precision and recall) for GPT-4, Llama 2, and Gemini Pro across different prompt settings.

Lang.	Class	P1		P2		P3		P4		P5	
		P.	R.	P.	R.	P.	R.	P.	R.	P.	R.
GPT-4											
BN	Cont	89.45	62.39	89.45	62.39	-	-	86.48	69.08	86.19	66.26
	Ent	86.13	54.45	86.13	54.45	-	-	79.80	62.23	71.69	56.71
	Neut	53.50	89.29	53.50	89.29	-	-	56.92	80.84	56.60	85.50
EN	Cont	92.45	89.40	-	-	92.87	89.64	86.56	93.35	93.26	88.68
	Ent	88.25	86.88	-	-	87.84	87.84	79.26	88.08	88.20	87.25
	Neut	80.02	83.90	-	-	80.79	83.59	81.23	65.81	79.49	84.25
HI	Cont	89.95	68.62	91.08	66.65	90.78	65.45	87.98	70.54	-	-
	Ent	83.00	60.24	83.12	57.49	84.60	51.32	80.02	61.62	-	-
	Neut	56.70	85.69	54.96	86.65	53.10	88.80	57.02	81.44	-	-
UR	Cont	67.54	79.88	82.67	64.85	84.57	62.04	-	-	79.22	68.50
	Ent	74.89	52.51	76.24	54.37	79.86	49.40	-	-	79.86	49.40
	Neut	56.28	62.81	53.33	80.12	51.27	84.49	-	-	51.27	84.49
Llama 2											
BN	Cont	06.61	06.20	14.95	06.75	-	-	15.28	08.47	04.82	03.01
	Ent	11.08	12.88	86.13	54.45	-	-	26.80	53.65	22.21	41.20
	Neut	97.01	87.27	98.57	29.56	-	-	99.17	44.12	99.41	51.90
EN	Cont	86.78	90.00	-	-	73.25	80.66	85.90	80.24	77.65	83.83
	Ent	52.84	95.81	-	-	49.05	81.98	48.87	87.01	50.69	83.77
	Neut	76.00	11.38	-	-	86.84	19.76	71.91	20.54	86.58	23.17
HI	Cont	11.88	11.80	0.60	0.60	19.10	17.54	0.89	0.89	-	-
	Ent	13.22	15.75	0.77	0.78	20.92	29.28	0.66	0.66	-	-
	Neut	99.56	81.26	100.00	99.22	98.68	67.25	99.45	99.45	-	-
UR	Cont	33.14	26.77	48.66	32.57	28.73	21.08	-	-	10.91	11.08
	Ent	30.50	59.88	33.17	72.93	28.56	54.79	-	-	11.44	14.13
	Neut	98.43	22.51	98.18	12.93	98.97	34.43	-	-	99.68	74.73
Gemini											
BN	Cont	63.32	78.47	64.60	74.11	-	-	61.35	78.77	68.01	69.26
	Ent	66.25	54.63	67.48	46.05	-	-	82.42	18.40	68.56	43.59
	Neut	57.12	53.49	52.59	61.54	-	-	48.58	72.46	51.41	69.16
EN	Cont	76.97	87.84	-	-	74.27	88.14	74.36	84.78	82.31	82.75
	Ent	73.79	71.32	-	-	73.53	75.51	72.78	29.94	88.20	87.25
	Neut	69.44	61.95	-	-	72.87	57.28	51.16	74.12	79.49	84.25
HI	Cont	68.03	72.87	69.40	69.94	63.55	79.34	63.83	62.99	-	-
	Ent	68.92	48.74	68.24	47.60	64.16	58.86	63.57	15.57	-	-
	Neut	52.45	64.07	52.59	68.08	57.65	48.08	43.96	77.72	-	-
UR	Cont	42.93	71.86	59.34	77.07	64.72	73.52	-	-	57.93	78.37
	Ent	42.06	13.95	63.64	41.94	67.00	40.60	-	-	66.70	41.50
	Neut	63.80	63.45	50.75	52.88	50.07	63.01	-	-	50.61	51.89

Table 7: Detailed Class-wise result for GPT-4, Llama 2, and Gemini Pro. Lang: Language, P: Precision, R: Recall, Cont: Contradiction, Ent: Entailment, Neut: Neutral.